

[Gayathri K 251801076-2025-AIDS](#) ▾**G2****Started on** Friday, 9 October 2026, 10:01 PM**State** Finished**Completed on** Friday, 9 October 2026, 10:06 PM**Time taken** 5 mins 8 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int counter = 0;
9      int i = 1;
10     counter++;
11
12     int s = 1;
13     counter++;
14
15     while (s <= n)
16     {
17         counter++;
18         i++;
19         counter++;
20         s += i;
21         counter++;
22     }
23
24     counter++;
25     printf("%d", counter);
26
27     return 0;
28
29
30 }
31
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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G2**Started on** Tuesday, 25 August 2026, 10:44 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 10:46 AM**Time taken** 2 mins 48 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, count = 0;
6
7      scanf("%d", &n);
8
9      if (n == 1)
10     {
11         count = 1;
12     }
13     else
14     {
15         count = 5 * n + 2;
16     }
17
18     printf("%d", count);
19
20     return 0;
21 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓

	Input	Expected	Got	
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Gayathri K 251801076-2025-AIDS](#) ▾**G2****Started on** Tuesday, 25 August 2026, 10:47 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 10:54 AM**Time taken** 7 mins 8 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int num, i, count = 0;
6
7      scanf("%d", &num);
8
9      i = 1;
10     while (1)
11     {
12         count++;
13         if (!(i <= num))
14             break;
15
16         count++;
17         if (num % i == 0)
18         {
19             count++;
20         }
21         i++;
22     }
23
24     printf("%d", count);
25     return 0;
26 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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G2**Started on** Friday, 9 October 2026, 10:12 PM**State** Finished**Completed on** Friday, 9 October 2026, 10:12 PM**Time taken** 32 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5  int n;
6  long long c = 0;
7
8
9  scanf("%d", &n);
10
11 c++;
12
13 for (int i = n / 2; ; i++)
14 {
15     c++;
16     if (!(i < n)) break;
17
18     for (int j = 1; ; j = 2 * j)
19     {
20         c++;
21         if (!(j < n)) break;
22
23         for (int k = 1; ; k = k * 2)
24         {
25             c++;
26             if (!(k < n)) break;
27
28             c++;
29         }
30     }
31 }
32
33 printf("%lld", c);
34 return 0;
35
36
37 }
38
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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G2**Started on** Friday, 9 October 2026, 10:13 PM**State** Finished**Completed on** Friday, 9 October 2026, 10:14 PM**Time taken** 41 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, rev = 0, remainder;
6      long long c = 2;
7
8
9      scanf("%d", &n);
10
11     while (1)
12     {
13         c++;
14         if (!(n != 0))
15             break;
16
17         remainder = n % 10;
18         c++;
19
20         rev = rev * 10 + remainder;
21         c++;
22
23         n /= 10;
24         c++;
25     }
26
27     printf("%lld", c);
28     return 0;
29
30
31 }
32
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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